

Non-stationary simulation study: what a mean-only MRTS augmentation can and cannot buy

July 2026 — interim report v3 (settings 1–12, the threshold-vs-symmetry disambiguation,
and the $K \geq 30$ extension)

Abstract

We evaluate the MRTS covariate extension of the Morris et al. (2017) skew- t space-time model on twelve simulated data-generating processes (DGPs) that place non-stationarity in different moments: the mean (settings 1, 2, 11, 12), the spatial dependence (settings 3–8), the tails (setting 9), and the skewness (setting 10). Three findings. (i) MRTS improves predictive scores *only* where the truth carries time-averaged mean structure, and this is provable in advance: an oracle calculation shows the relative Brier score has a floor $R^* \approx 1$ for nine of eleven settings, so their observed nulls are properties of the design, not evidence about MRTS. (ii) Where mean structure exists, the energy- and variogram-score gains are carried *entirely* by the two thresholded symmetric- t methods. A disambiguating pair of methods (symmetric- t *without* threshold, $n = 10$) attributes this cleanly to the *threshold*, not the symmetry — and the absolute scores reframe it: POT censoring without a mean basis damages the predictive field so badly (baseline energy $2.6 \times$ – $16 \times$ worse) that MRTS is a partial *rescue*, not a synergy; the unthresholded methods never needed rescuing and remain better in absolute terms at every K . (iii) The Brier score is close to blind to mean recovery and can even deteriorate with K : in the strong-mean setting 11 a fit that recovers every parameter of the DGP at $K = 25$ has a *worse* Brier score than the badly misspecified $K = 0$ baseline. The rich-mean setting 12, run out to $K = 50$, resolves this: the deterioration is confined to intermediate K (where the mean basis is still too coarse), and once the basis is fine enough ($K \geq 30$ – 40) the Brier score improves by 7–12%, consistent with the extrapolation-error mechanism. The discriminating check passes: the simple-surface setting 11, extended to the same $K = 50$, never turns positive — its Brier ratio only decays back to parity ($1.09 \rightarrow 1.00$) — so the reversal requires mean structure that only a fine basis can resolve, not merely a large K .

1 Design

Data-generating process. Settings 1–12 share the skew- t -1 core on $n_s = 144$ sites drawn uniformly on $[0, 10]^2$ (settings 13–21 replace it with a Gaussian process; see below):

$$Y_t(\mathbf{s}) = 10 + \lambda \sigma_t |z_t| + \sigma_t v_t(\mathbf{s}) + [\text{structure}]_k, \quad t = 1, \dots, 50, \quad (1)$$

with $\lambda = 3$, $z_t \stackrel{iid}{\sim} N(0, 1)$ half-normal, $v_t \sim N(0, C)$ exponential correlation ($\rho = 1$, $\gamma = 0.9$), $\tau_t \sim \text{Ga}(3/2, 4)$, $\sigma_t = \tau_t^{-1/2}$ (marginal t_3), i.i.d. in time. The settings differ only in the structure term—or, for the *replace- C* settings, in the error correlation itself (Table 1). Table 2 gives the resulting anatomy of each DGP: the marginal law of $Y_t(\mathbf{s})$, the spatial dependence of the error process, and the time-averaged true mean surface $\bar{\mu}(\mathbf{s})$ (the target of any pooled- β mean basis). Settings 13–21 form a *non- t extension*: they drop the skew- t generator entirely in favour of a

nonlinear mean surface (sd 11) with Gaussian-GP errors ($\sigma_g = 4$, same C_{exp}), so that the mean channel is tested without the crutch of a correctly specified error law at large K . Their fitting is in progress; the design tables list all 21 settings, while the score tables report settings that have been fitted.

Table 1: The DGP settings and the number of simulation replicates n (independent datasets) fitted per setting. Settings 1–12 share the skew- t core; settings 13–21 are the non- t extension (Gaussian-GP errors, $\sigma_g = 4$; nonlinear mean surface orthogonalised against $[1, s_1, s_2]$ and standardised to sd = 11). Routes: *mean* = added to the fixed mean; *additive* = mean-zero random field added to the draw; *replace C* = non-stationary correlation handed to the generator (unit diagonal, so the *marginal* skew- t law is unchanged — verified by KS tests, $p \in [0.075, 0.864]$); *skew term* = spatially varying λ .

ID	surf_type	non-stationarity	route	n
1	invariant	mean, 2 cosine bumps, sd = 3.7	mean	10
2	varying	mean, time-varying weights	mean	10 ^a
3	ns_dependence	dependence, low-rank cosine RE	additive	10
4	ps_cov	dependence, Paciorek–Schervish	replace C	3
5	ps_add	dependence, same PS field	additive	3
6	covdep_cov	dependence, $\rho(\mathbf{s}) = e^{0.7f_1(\mathbf{s})}$	replace C	3
7	covdep_add	dependence, same field	additive	3
8	fuentes_cov	dependence, 4-regime mixture	replace C	3
9	gh_marginal	tails, Tukey g -and- h , $g(\mathbf{s}), h(\mathbf{s})$	additive	3
10	lambda_varying	skewness, $\lambda(\mathbf{s}) = 3 + 3f_2(\mathbf{s})$	skew term	3
11	invariant_strong	mean, 3× setting 1, sd = 11	mean	3
12	mrts_multibump	mean, 6 Gaussian bumps, sd = 11	mean	3
<i>Non-t extension (Gaussian-GP errors; skew-t generator dropped)</i> ^b				
13	franke	mean, Franke (multiscale)	mean	3
14	ridge_curved	mean, curved Gaussian ridge	mean	3
15	annulus	mean, Ricker ring ($r_0=3, w=1.5$)	mean	3
16	sigmoid_front	mean, tanh arc front	mean	3
17	gp_mean	mean, GP draw (per dataset, fixed in t)	mean	3
18	franke_sas	mean = setting 13; sinh-arcsinh error	mean	3
19	poly2	mean, bowl + saddle (2nd-order poly)	mean	3
20	transform_mix	mean, log + exp-decay + ratio terms	mean	3
21	nonsmooth	mean, hinge + $ \cdot $ (C^0 creases)	mean	3

^a setting 2 was fitted at $K = 0$ only (no MRTS sweep); it is excluded from the score tables. ^b settings 13–21 use the grid $K_{\text{MRTS}} \in \{0, 30, 40, 50\}$; fitting is in progress (setting 13 complete at time of writing).

Fitted methods and grid. Five baseline methods are fitted to every dataset: (1) Gaussian; (2) skew- t , $K_{\text{knots}} = 1$; (3) symmetric- t , $K_{\text{knots}} = 1$, POT threshold at $q(0.80)$; (4) skew- t , $K_{\text{knots}} = 5$; (5) symmetric- t , $K_{\text{knots}} = 5$, threshold $q(0.80)$. Each is run across MRTS design widths $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$, where $K_{\text{MRTS}} = 0$ denotes the baseline design $[1, s_1, s_2]$ and $K_{\text{MRTS}} = K$ replaces the design with the K -column `autoFRK::mrts` basis built on the *training* sites only (column 1 = intercept, 2–3 linear, 4: K thin-plate-spline eigenfunctions; $K = 3$ spans the baseline). MCMC: 20,000 iterations, 10,000 burn-in. The random seed is shared across K_{MRTS} within each (method, dataset), so all K -comparisons are *paired*. The grid is being extended by $K_{\text{MRTS}} \in \{30, 40, 50\}$ setting by setting (complete for settings 1, 4, 11, 12 at the time of writing; Table 6 shows exactly which cells exist).

Scoring. The last 44 of the 144 sites are held out entirely; all scores are computed out-of-sample on those sites. With $\hat{F}$ the posterior predictive law:

$$\text{BS} = \frac{1}{11} \sum_p \frac{1}{44 \cdot 50} \sum_{i,t} (\hat{P}(Y_t(\mathbf{s}_i^*) > u_p) - \mathbb{I}\{y_t(\mathbf{s}_i^*) > u_p\})^2, \quad p \in \{0.90, \dots, 0.995\}, \quad (2)$$

$$\text{ES} = \frac{1}{50} \sum_t \left[\mathbb{E} \|\mathbf{X}_t - \mathbf{y}_t\| - \frac{1}{2} \mathbb{E} \|\mathbf{X}_t - \mathbf{X}'_t\| \right], \quad \mathbf{X}_t, \mathbf{X}'_t \sim \hat{F}_t \text{ (44-dim)}, \quad (3)$$

$$\text{VS} = \frac{1}{50} \sum_t \sum_{i < j} (|y_{it} - y_{jt}|^{1/2} - \mathbb{E} |X_{it} - X_{jt}|^{1/2})^2. \quad (4)$$

The Brier score is a per-cell *marginal* score and cannot see spatial dependence; the energy score is the multivariate CRPS (reduces to CRPS at $d = 1$) and is location-sensitive; the variogram score ($p = 1/2$) targets pairwise dependence. Every result below is the *relative* score $R_{m,K,d} = S_{m,K,d}/S_{m,0,d}$ against the same method’s own $K = 0$ fit, averaged over datasets; $R < 1$ means MRTS helps.

2 Results

Tables 3–5 report, per setting and method, the best (smallest) $\bar{R}_{m,K}$ over $K > 0$, with the minimising K in parentheses. Asterisks mark $|\bar{R} - 1| > 2\widehat{\text{SE}}$ (SE over datasets; with $n = 3$ this is indicative only). Table 6 gives the full setting $\times K$ profile of the relative Brier score, method by method.

2.1 Finding 1: gains appear only where the truth has mean structure — and this is provable a priori

The only large entries in any table are settings 1, 11 and 12 (mean structure), methods 3 and 5: energy and variogram reductions of 34–42% (setting 1), 73–80% (setting 11) and 58% (setting 12). Every dependence, tails and skewness setting is flat to within a few percent, in *all three* scores.

This dichotomy can be established without MCMC. Because the MRTS coefficient β is pooled over time, the most any mean basis can deliver is the time-averaged true mean $\bar{\mu}(\mathbf{s})$. Define the oracle relative Brier

$$R^* = \frac{\text{BS}(\bar{\mu})}{\text{BS}(\text{spatial constant})}, \quad (5)$$

both evaluated under the *true* skew- t predictive law. R^* is a floor: no K , sample size or sampler can push the relative Brier below it. Table 7 shows $R^* \approx 1$ for nine of eleven settings: their Brier nulls are design properties, not evidence about MRTS. Setting 10 is the striking case: $R^* = 1.12 > 1$ — supplying the *true* mean makes the Brier score *worse*, because the mean shift induced by $\lambda(\mathbf{s})$ is a symptom of a skewness change that the constant- λ model cannot represent; correcting the symptom while the disease remains actively misleads the tail probabilities.

2.2 Finding 2: the gains are carried entirely by the thresholded methods

In Table 3, setting 11 (mean sd = 11), the two POT-thresholded symmetric- t methods reduce the energy score by 73–80%, while the Gaussian and both skew- t methods stay within 2% of their $K = 0$ baselines — despite those same skew- t fits recovering the mean surface almost exactly (recovery RMSE $14.0 \rightarrow 0.66$ for method 2 at $K = 25$) and recovering every DGP parameter (Section 2.3). Table 8 shows the full K profile.

Methods 3/5 are simultaneously (a) POT-thresholded and (b) symmetric- t , so a disambiguating pair was added: methods 11/12, *identical to methods 3/5 except* threshold = 0, fitted on setting 1 with $n = 10$ (Table 9). The verdict is unambiguous: the unthresholded symmetric- t methods are *flat* (energy ratio 0.99 at every K , indistinguishable from the skew- t reference), while their thresholded twins gain 35–40%. **The threshold, not the symmetry, is the driver.**

The absolute column reframes the finding. The thresholded baselines are *catastrophically* worse to begin with: censoring everything below $q(0.80)$ leaves the constant-mean fit with almost no information about the spatial level of the field, and its out-of-sample predictive field degrades by a factor 2.6 (K1) to 16 (K5) in energy score. MRTS repairs the mean that the censored likelihood cannot learn — a partial *rescue*, not a synergy: even at the best K , method 3 reaches $21.8 \times 0.579 \approx 12.6$, still well above the 8.5 that methods 2/11 achieve *without any MRTS at all*. In absolute terms the unthresholded methods win at every K ; the correct reading of Finding 2 is therefore “*only the thresholded methods have room to be helped*,” and the practical implication is for POT-style threshold models specifically: if the mean is not modelled well, censored fitting amplifies the damage, and a mean basis buys back part of it.

2.3 Finding 3: the Brier score is nearly blind to mean recovery, and can punish it

Setting 11, method 2 (skew- t K1), dataset 1, posterior means (truth: $\lambda = 3$, $\mathbb{E}[\sigma] = 2.26$, $\rho = 1$, $\gamma = 0.9$):

K_{MRTS}	recovery RMSE	$\hat{\lambda}$	$\mathbb{E}[\hat{\sigma}]$	$\hat{\rho}$	$\hat{\gamma}$
0	14.01	0.15	11.16	2.04	0.99
10	2.51	2.04	3.01	1.21	0.91
25	0.66	3.18	1.94	0.96	0.94

At $K = 0$ the model cannot represent the sd-11 mean surface and absorbs the spatial variance into an inflated σ ($5\times$ truth) with λ crushed to zero. At $K = 25$ *every* parameter is recovered. Yet the $K = 25$ Brier score is *worse* than $K = 0$ by 8.6% (Table 8), and the same reversal holds for methods 1, 4, 5. A correctly specified model loses to a badly misspecified one on this score. The working hypothesis — not yet verified — is an out-of-sample interaction: the sharpened predictive distribution ($\hat{\sigma}$ correct instead of inflated) exposes the residual error of the MRTS mean *extrapolated* to the held-out sites, whereas the misspecified $K = 0$ fit hides that error inside an accidentally conservative (inflated- σ) predictive distribution.

Setting 12 resolves the anomaly. The rich-mean setting 12 (six bumps, sd 11) was run out to $K_{\text{MRTS}} = 50$, where the noise-free MRTS projection error of its surface finally becomes small (0.71 at $K = 5$, 0.13 at $K = 25$, 0.057 at $K = 50$). The Brier profile follows exactly the pattern the extrapolation mechanism predicts: flat-to-worse at intermediate K (ratios 1.00–1.05 for $K \leq 25$, where the basis is still too coarse and the residual mean error is exposed by the sharpened predictive), then *improving* once the basis is fine enough:

relative Brier, setting 12	$K=15$	$K=25$	$K=30$	$K=50$	recovery at $K=50$
Gaussian	1.019	1.045	0.988	0.930	0.512
Skew- t K1	1.022	1.020	0.980	0.925	0.073
Sym- t K1 thr.	0.899	0.918	0.892	0.880	0.422
Skew- t K5	0.976	1.022	0.965	0.913	0.268

(Method 5 is excluded: its exponential-fallback instability blows up one of the three datasets at large K , ratio 2.7.) The $K=50$ gains of 7–12% are real but far above the oracle floor $R^* = 0.495$ — the mean channel is recovered (skew- t K1 recovery ratio 0.073), yet most of the theoretically available Brier headroom is spent compensating the sharpness–extrapolation trade-off. Practically: *for judging mean-structure recovery, the Brier score remains the wrong instrument — it responds only when the basis is fine enough to make the residual extrapolation error negligible; energy/variogram scores (or the recovery RMSE directly) are the informative ones.*

The large- K limit discriminates the mechanism. If the reversal were simply a matter of “ K large enough,” setting 11 — whose 2-bump surface the basis captures by $K \approx 10$ — should also turn positive by $K = 50$. It does not (Table 8, rightmost columns): the mid- K deterioration decays monotonically (1.16 at $K=12$ to 1.01–1.00 at $K=50$ for the skew- t and Gaussian methods) but never crosses below parity, leaving the oracle headroom $R^* = 0.439$ entirely unused. The weak-signal setting 1 behaves the same way at large K (skew- t K1: 0.985 at $K=25$, 0.978 at $K=50$ — smooth, small, no regime change). Same K grid, same machinery, opposite outcome to setting 12: the Brier reversal requires mean structure that *only* a fine basis can resolve; extra columns on an already-resolved surface buy nothing and merely let the sharpness–extrapolation penalty fade.

3 Caveats

- Settings 4–11 are pilots with $n = 3$; settings 1 and 3 have $n = 10$. Starred cells with $n = 3$ should be read as indicative. The small- n *best-over- K* statistic also has a selection bias toward values < 1 ; flat profiles (e.g. Sym- t K5 rows in dependence settings, best ≈ 0.72 –0.85 with non-monotone K profiles) are consistent with noise plus this selection effect.
- Method 5 (Sym- t K5 T) occasionally triggers its exponential-covariance fallback and produces isolated outlier fits (e.g. setting 4, $K = 25$: energy ratio 15.8 in one dataset). Best-over- K cells are robust to this; per- K curves at $n = 3$ are not.
- All K comparisons are paired (shared seeds across K), so within-method ratios are considerably more precise than the between-method comparisons.
- The recovery RMSE (not tabulated here) has a known bimodality on flat-mean settings caused by the intercept- λ ridge; see `nonsta_settings_design.md`.

4 The setting-12 showcase: why a large K is worth having

Setting 12’s six-bump surface was designed so that the MRTS approximation improves *gradually* with K (noise-free projection error 0.71 at $K=5$, 0.19 at $K=15$, 0.057 at $K=50$), in contrast to the 2-bump settings which the basis captures by $K=5$. The fitted results track that curve: skew- t K1 recovery ratios are 0.93 ($K=5$) $\rightarrow$ 0.49 ($K=10$) $\rightarrow$ 0.14 ($K=25$) $\rightarrow$ 0.073 ($K=50$) — the extended elbow, realised in MCMC. Two by-products: (a) the Brier anomaly resolution above; (b) a resolution ceiling — a higher-frequency mean component (3 cycles per axis) is *unrecoverable* on the 100-train/44-test geometry (its projection error *rises* with K), i.e. MRTS’s usable resolution is capped by site density, and the six-bump surface sits just inside that cap.

5 In progress

1. **Non- t extension (settings 13–21):** all settings so far share the skew- t core, so at large K the fitted skew- t is *correctly specified* — a confound for every “MRTS helps/hurts” conclusion. The extension drops the skew- t generator entirely: nine nonlinear mean surfaces (Franke, curved ridge, annulus, sigmoid front, GP draw, polynomial, transform mixtures, non-smooth hinges; orthogonalised against the $K = 0$ design and standardised to sd 11) with Gaussian-GP errors, calibrated so that $R^* \approx 0.45$ throughout. Data generation in `setup_nonsta.R` $\rightarrow$ `simdata_nonsta_ext.RData`.
2. **n upgrade:** settings 4–12 are $n = 3$ pilots; the headline contrasts (11, 12, and the disambiguation) merit $n = 10$.

Table 2: Anatomy of the DGPs: marginal law of $Y_t(\mathbf{s})$, spatial dependence of the error process, and time-averaged true mean $\bar{\mu}(\mathbf{s})$ (its spatial sd in parentheses; per-dataset values for random means). $C_{\text{exp}}(h) = 0.9 e^{-h}$ is the shared stationary exponential correlation (Matérn $\nu = \frac{1}{2}$, $\rho = 1$, unit diagonal, i.e. a 10% nugget); “skew- t ” is the t_3 core of (1) with $\lambda = 3$; $*$ denotes convolution with an independent additive field ($\sigma_{\text{add}} = 3$); f_1, f_2 are the two cosine bumps. For the non- t block (13–21) the error is $\sigma_g \varepsilon$ with $\varepsilon \sim N(0, C_{\text{exp}})$, $\sigma_g = 4$ (so the marginal is Gaussian, sd 4), and every mean surface is orthogonalised against $[1, s_1, s_2]$ and rescaled to sd = 11.

ID	surf_type	marginal law	spatial dependence	mean structure $\bar{\mu}(\mathbf{s})$
1	invariant	skew- t	C_{exp} , stationary	$10 + 5f_1 + 3f_2$ (3.7)
2	varying	skew- t	C_{exp} , stationary	$10 + w_{1t}f_1 + w_{2t}f_2$, (w_{1t}, w_{2t}) $\sim$ $N(0, \text{diag}(25, 9))$ per day; $\bar{\mu} \approx 10$ (0.4)
3	ns_dependence	skew- $t * \text{Gaussian}$	$C_{\text{exp}} + \text{low-rank cosine}$ $\sum_j \tau_j^2 \phi_j(\mathbf{s}) \phi_j(\mathbf{s}')$; non- stationary, negative at long range	10 (RE mean-zero) (0.4)
4	ps_cov	skew- t (unchanged)	Paciorek–Schervish locally anisotropic kernel; scale, as- pect and rotation each vary across the domain	10 exactly (0)
5	ps_add	skew- $t * \text{Gaussian}$	$C_{\text{exp}} + \sigma_{\text{add}}^2 C_{\text{PS}}$, additive	10 (RE mean-zero) (0.4)
6	covdep_cov	skew- t (unchanged)	isotropic PS kernel with lo- cal range $\rho(\mathbf{s}) = e^{0.7f_1(\mathbf{s})} \in$ $[0.5, 2.0]$	10 exactly (0)
7	covdep_add	skew- $t * \text{Gaussian}$	$C_{\text{exp}} + \sigma_{\text{add}}^2 C_{\rho(\mathbf{s})}$, additive	10 (RE mean-zero) (0.4)
8	fuentes_cov	skew- t (unchanged)	Fuentes 4-regime weighted GP mixture, ranges 0.4–4	10 exactly (0)
9	gh_marginal	skew- $t * \text{std. Tukey } g\text{-}$ $\text{and-}h$; $g(\mathbf{s}) = 0.8f_1$, $h(\mathbf{s}) = 0.12(1+f_2)$	C_{exp} in both components (transform perturbs correla- tion only mildly)	10 (standardised mean- zero) (0.4)
10	lambda_varying	skew- t with $\lambda(\mathbf{s}) = 3 +$ $3f_2 \in [0, 6]$	C_{exp} , stationary	induced: $10 + \lambda(\mathbf{s}) \bar{z}$, $\bar{z} =$ $\frac{1}{T} \sum_t \sigma_t z_t $ (2.8)
11	invariant_strong	skew- t	C_{exp} , stationary	$10 + 15f_1 + 9f_2$ (11.0)
12	mrts_multibump	skew- t	C_{exp} , stationary	$10 + \text{six alternating-sign}$ $\text{Gaussian bumps (width}$ $1.6)$ (11.0)
<i>Non-t extension: Gaussian-GP errors, $\sigma_g = 4$, mean surface sd 11</i>				
13	franke	Gaussian	C_{exp} , stationary	$10 + \text{Franke's function (2}$ $\text{large} + 1 \text{ medium bump, 1}$ fine dip) (11.0)
14	ridge_curved	Gaussian	C_{exp} , stationary	$10 + \text{curved Gaussian}$ $\text{ridge } \exp[-(s'_2 - 0.5 -$ $0.25 \sin 2\pi s'_1)^2/2 - 0.12^2]$ (11.0)
15	annulus	Gaussian	C_{exp} , stationary	$10 + \text{Ricker ring } \psi((r -$ $3)/1.5), r = \ \mathbf{s} - (5, 5)\ $ (11.0)
16	sigmoid_front	Gaussian	C_{exp} , stationary	$10 + \tanh[(\ \mathbf{s} - (2, 2)\ -$ $4.5)/1.0]$ arc front (11.0)
17	gp_mean	Gaussian	C_{exp} , stationary	$10 + \text{GP draw (Matérn}$ $\nu=1.5, \text{ range } 2)$, redrawn per dataset, fixed in t (11.0)
18	franke_sas	$\sinh\text{-arcsinh}$ ($\epsilon=0.8$, $\delta=1$), right-skewed	C_{exp} , stationary	$10 + \text{Franke (identical to}$ setting 13) (11.0)
19	poly2	Gaussian	C_{exp} , stationary	$10 + \text{2nd-order polynomial}$ (bowl + saddle) (11.0)
20	transform_mix	Gaussian	C_{exp} , stationary	$10 + \log(1+s_1) + e^{-s_1/2} +$ $s_1/(1+s_2)$ (11.0)
21	nonsmooth	Gaussian	C_{exp} , stationary	$10 + \text{hinge}(s_1-5) + s_2 - 4 $ (C^0 creases) (11.0)

Table 3: **Energy score**, best over $K > 0$ relative to the same method's $K = 0$ (minimising K in parentheses). Values < 1 : MRTS helps; bold: $\bar{R} < 0.75$.

ID	non-stationarity	n	Gaussian	Skew- t K1	Sym- t K1 T	Skew- t K5	Sym- t K5 T
1	mean (2 bumps, sd 3.7)	10	0.994* ⁽⁴⁰⁾	0.980* ⁽⁴⁰⁾	0.579 * ⁽²⁰⁾	0.983* ⁽⁵⁰⁾	0.659 * ⁽¹²⁾
3	dependence (low-rank)	10	1.000 ⁽⁵⁾	1.000 ⁽³⁾	0.993 ⁽⁵⁾	1.000 ⁽¹⁵⁾	0.911 ⁽³⁾
4	dependence (PS, repl. C)	3	0.999* ⁽⁵⁾	1.000 ⁽³⁰⁾	0.889* ⁽⁴⁰⁾	0.996* ⁽³⁾	0.846 ⁽¹⁵⁾
5	dependence (PS, additive)	3	0.999 ⁽¹²⁾	0.999* ⁽⁸⁾	0.953 ⁽¹⁵⁾	0.998* ⁽¹⁰⁾	0.955 ⁽¹⁵⁾
6	dependence ($\rho(\mathbf{s})$, repl. C)	3	1.000 ⁽¹⁰⁾	0.999 ⁽¹⁰⁾	0.861* ⁽²⁰⁾	0.997* ⁽¹²⁾	0.908 ⁽⁵⁰⁾
7	dependence ($\rho(\mathbf{s})$, additive)	3	1.000 ⁽²⁵⁾	0.999 ⁽³⁰⁾	0.985 ⁽³⁰⁾	0.995 ⁽²⁵⁾	0.926 ⁽¹⁵⁾
8	dependence (Fuentes, repl. C)	3	1.000 ⁽¹⁰⁾	1.000 ⁽³⁾	0.848 ⁽²⁵⁾	0.998 ⁽⁵⁾	0.695 ⁽³⁰⁾
9	tails (g -and- h)	3	0.999* ⁽⁵⁾	1.000 ⁽¹⁰⁾	0.887 ⁽⁵⁾	0.986 ⁽²⁵⁾	0.942 ⁽⁵⁾
10	skewness ($\lambda(\mathbf{s})$)	3	0.993* ⁽⁵⁰⁾	1.000 ⁽⁵⁰⁾	0.975 ⁽⁵⁾	0.995 ⁽³⁰⁾	0.938* ⁽⁵⁾
11	mean (2 bumps, sd 11)	3	0.976* ⁽⁵⁰⁾	0.965* ⁽⁵⁰⁾	0.199 * ⁽²⁰⁾	0.961* ⁽⁵⁰⁾	0.267 * ⁽²⁰⁾
12	mean (6 bumps, sd 11)	3	0.960* ⁽⁵⁰⁾	0.953* ⁽⁵⁰⁾	0.421 * ⁽⁴⁰⁾	0.952* ⁽⁵⁰⁾	0.846 ⁽¹⁵⁾
<i>Non-t extension (Gaussian-GP errors; skew-t is misspecified)</i>							
13	mean, Franke (multiscale)	3	0.944* ⁽⁵⁰⁾	0.945* ⁽⁵⁰⁾	1.198 ⁽⁵⁰⁾	0.942* ⁽⁵⁰⁾	1.179 ⁽⁴⁰⁾

Table 4: **Brier score** (mean over the 11 exceedance levels), best over $K > 0$ relative to own $K = 0$. Bold: $\bar{R} < 0.75$.

ID	non-stationarity	n	Gaussian	Skew- t K1	Sym- t K1 T	Skew- t K5	Sym- t K5 T
1	mean (2 bumps, sd 3.7)	10	1.000 ⁽³⁾	0.978* ⁽⁵⁰⁾	0.993* ⁽³⁰⁾	0.984* ⁽⁵⁰⁾	0.971 ⁽³⁾
3	dependence (low-rank)	10	1.000 ⁽⁵⁾	0.999 ⁽⁵⁾	1.000 ⁽³⁾	0.996 ⁽⁴⁰⁾	0.993 ⁽¹²⁾
4	dependence (PS, repl. C)	3	0.998 ⁽⁵⁰⁾	0.998* ⁽³⁰⁾	0.998* ⁽³⁰⁾	0.983* ⁽⁵⁰⁾	0.940* ⁽²⁰⁾
5	dependence (PS, additive)	3	1.000 ⁽³⁾	0.999 ⁽¹²⁾	1.000 ⁽⁸⁾	0.990 ⁽¹⁵⁾	0.973 ⁽⁵⁾
6	dependence ($\rho(\mathbf{s})$, repl. C)	3	0.998 ⁽⁵⁰⁾	0.999* ⁽⁵⁾	0.997* ⁽⁴⁰⁾	0.983* ⁽¹²⁾	0.970 ⁽¹⁰⁾
7	dependence ($\rho(\mathbf{s})$, additive)	3	1.000 ⁽¹⁵⁾	0.999 ⁽¹⁰⁾	0.998 ⁽¹⁵⁾	0.990 ⁽⁵⁾	0.968 ⁽¹⁵⁾
8	dependence (Fuentes, repl. C)	3	1.000* ⁽³⁾	1.000 ⁽³⁾	0.999 ⁽³⁾	0.997 ⁽¹²⁾	0.962 ⁽¹²⁾
9	tails (g -and- h)	3	1.000 ⁽¹²⁾	1.000 ⁽¹⁵⁾	0.998 ⁽⁴⁰⁾	0.996 ⁽¹⁰⁾	0.967 ⁽¹²⁾
10	skewness ($\lambda(\mathbf{s})$)	3	0.988 ⁽⁵⁰⁾	0.998 ⁽⁵⁰⁾	0.949* ⁽⁵⁰⁾	0.988 ⁽³⁰⁾	0.972 ⁽³⁰⁾
11	mean (2 bumps, sd 11)	3	1.000 ⁽³⁾	1.000 ⁽³⁾	0.952* ⁽³⁰⁾	1.004 ⁽³⁾	0.997 ⁽³⁾
12	mean (6 bumps, sd 11)	3	0.930 ⁽⁵⁰⁾	0.925* ⁽⁵⁰⁾	0.880* ⁽⁵⁰⁾	0.913* ⁽⁵⁰⁾	0.884* ⁽¹⁰⁾
<i>Non-t extension (Gaussian-GP errors; skew-t is misspecified)</i>							
13	mean, Franke (multiscale)	3	0.974* ⁽⁵⁰⁾	0.972* ⁽⁵⁰⁾	0.956 ⁽⁵⁰⁾	0.950* ⁽⁵⁰⁾	0.936* ⁽³⁰⁾

Table 5: **Variogram score** ($p = 1/2$), best over $K > 0$ relative to own $K = 0$. Bold: $\bar{R} < 0.75$.

ID	non-stationarity	n	Gaussian	Skew- t K1	Sym- t K1 T	Skew- t K5	Sym- t K5 T
1	mean (2 bumps, sd 3.7)	10	0.994* ₍₄₀₎	0.975* ₍₄₀₎	0.617* ₍₁₅₎	0.981* ₍₅₀₎	0.712 ₍₁₂₎
3	dependence (low-rank)	10	1.000 ₍₅₎	1.000 ₍₃₎	0.984 ₍₅₎	0.999 ₍₂₅₎	0.894 ₍₃₎
4	dependence (PS, repl. C)	3	0.999* ₍₁₀₎	1.000 ₍₁₅₎	0.988 ₍₁₅₎	0.997 ₍₁₀₎	0.990 ₍₃₎
5	dependence (PS, additive)	3	0.999* ₍₁₂₎	0.999* ₍₈₎	0.963 ₍₅₀₎	0.998 ₍₁₅₎	0.986 ₍₁₅₎
6	dependence ($\rho(\mathbf{s})$, repl. C)	3	0.999 ₍₁₀₎	0.999 ₍₅₀₎	0.920* ₍₄₀₎	0.997 ₍₁₂₎	0.984 ₍₁₀₎
7	dependence ($\rho(\mathbf{s})$, additive)	3	1.000 ₍₁₀₎	1.000 ₍₃₀₎	0.995 ₍₃₎	0.995 ₍₅₎	0.960 ₍₁₅₎
8	dependence (Fuentes, repl. C)	3	1.000 ₍₈₎	1.000 ₍₃₎	0.887 ₍₄₀₎	1.000 ₍₃₎	0.847 ₍₅₎
9	tails (g -and- h)	3	1.000 ₍₃₎	1.000 ₍₅₎	0.865 ₍₅₎	0.982 ₍₂₅₎	0.947 ₍₅₎
10	skewness ($\lambda(\mathbf{s})$)	3	0.990 ₍₄₀₎	0.999 ₍₃₀₎	1.000 ₍₃₎	0.993 ₍₃₀₎	0.905* ₍₁₀₎
11	mean (2 bumps, sd 11)	3	0.963* ₍₅₀₎	0.943* ₍₅₀₎	0.201* ₍₁₀₎	0.935* ₍₅₀₎	0.244* ₍₂₀₎
12	mean (6 bumps, sd 11)	3	0.938* ₍₅₀₎	0.911* ₍₅₀₎	0.354* ₍₄₀₎	0.915* ₍₅₀₎	0.909 ₍₁₅₎
<i>Non-t extension (Gaussian-GP errors; skew-t is misspecified)</i>							
13	mean, Franke (multiscale)	3	0.912* ₍₅₀₎	0.913* ₍₅₀₎	1.164* ₍₃₀₎	0.912* ₍₅₀₎	1.085 ₍₃₀₎

Table 6: **Relative Brier score at every K_{MRTS}** (vs the same method’s own $K = 0$, mean over datasets; * marks $|\bar{R} - 1| > 2\widehat{\text{SE}}$). Setting 2 is excluded ($K = 0$ only); a dash means that K was not fitted for that setting (settings 13–21 use only $K \in \{30, 40, 50\}$). Isolated large values of Sym- t K5 T are the exponential-fallback outliers discussed in the Caveats and are reported as-is.

ID	n	$K=3$	$K=5$	$K=8$	$K=10$	$K=12$	$K=15$	$K=20$	$K=25$	$K=30$	$K=40$	$K=50$
<i>Gaussian</i>												
1	10	1.000	1.016*	1.023*	1.022*	1.019*	1.016*	1.009*	1.008*	1.005	1.003	1.003
3	10	1.000	1.000	1.001*	1.001*	1.000	1.000	1.000	1.001	1.001	1.001*	1.002*
4	3	1.000	0.999*	0.999*	1.000	1.000	1.000	0.999	0.999	0.999	0.999	0.998
5	3	1.000	1.001	1.001	1.000	1.001*	1.000	1.001*	1.002*	1.002*	1.001*	1.004*
6	3	1.000	1.000	1.000	1.000	1.000	1.000	0.999	1.000	0.999	0.998*	0.998
7	3	1.000	1.000	1.000	1.000	1.000	1.000	1.002*	1.001	1.000	1.000	1.000
8	3	1.000*	1.001	1.001	1.001	1.000	1.000	1.002	1.002	1.002	1.003	1.004*
9	3	1.000*	1.000	1.000	1.000*	1.000	1.000	1.000	1.001	1.001	1.002	1.001
10	3	1.000	1.003	1.008	1.004	1.000	0.997	0.995	0.995	0.989	0.988	0.988
11	3	1.000	1.059*	1.078*	1.101*	1.114*	1.095*	1.069*	1.051*	1.022	1.001	1.000
12	3	1.000	1.003*	1.000	0.993	0.980	1.019	1.049	1.045	0.988	0.947	0.930
		non- t ext. (Gaussian-GP)										
13	3	–	–	–	–	–	–	–	–	1.025	0.999	0.974*
<i>Skew-t K1</i>												
1	10	1.000	0.996	1.006	1.006	1.001	0.996	0.987*	0.985*	0.981*	0.978*	0.978*
3	10	1.000	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
4	3	1.000*	1.000	0.999	1.000	1.000	1.000	0.999	0.999	0.998*	0.999	0.999
5	3	1.000	1.000	0.999	1.000	0.999	1.000	1.000	1.001	1.001	1.001	1.002
6	3	1.000	0.999*	0.999*	1.000	1.001*	0.999	1.000	1.000	1.000	0.999	1.000
7	3	1.000	1.000	1.001	0.999	1.000	1.000	1.001	1.000	1.000	1.001	1.001
8	3	1.000	1.000	1.001	1.000	1.001	1.001*	1.001	1.000	1.001	1.001*	1.002*
9	3	1.001	1.001	1.001*	1.001	1.001	1.000	1.001	1.001	1.001	1.001	1.001
10	3	0.999	1.003	1.008*	1.008	1.005	1.003	1.002	1.001	0.999	0.998	0.998
11	3	1.000	1.052*	1.074*	1.103*	1.163*	1.138*	1.110*	1.086*	1.048	1.017	1.014
12	3	1.000	1.005*	0.999	0.992	0.965	1.022	1.024	1.020	0.980	0.946	0.925*
		non- t ext. (Gaussian-GP)										
13	3	–	–	–	–	–	–	–	–	1.024	0.998	0.972*
<i>Sym-t K1 T</i>												
1	10	1.000	1.003	1.006	1.008*	1.003	1.001	0.996	0.994*	0.993*	0.993*	0.994*
3	10	1.000	1.000	1.001	1.001	1.001	1.001	1.001	1.001	1.002	1.002	1.004*
4	3	1.000	0.998*	0.999	0.999	1.000	0.999	1.000	0.999	0.998*	1.000	0.999
5	3	1.000*	1.000	1.000	1.000	1.000	1.001	1.000	1.001	1.002*	1.004	1.007
6	3	1.000	1.002*	1.000	1.000	1.001	0.999*	0.999	1.000	1.001	0.997*	1.000
7	3	1.000*	1.000	1.000	1.000	1.000	0.998	1.001	1.000	1.000	0.999	1.001
8	3	0.999	0.999	0.999	1.002	1.000	1.001	1.003	1.008	1.009	1.012	1.010*
9	3	1.000	1.001	1.001	1.000	1.001	1.001*	1.000	1.001*	1.002	0.998	1.000
10	3	0.999	0.993*	0.980*	0.985	0.975	0.970	0.963*	0.957*	0.949*	0.954*	0.949*
11	3	0.999	1.016	1.021	1.039*	1.008	0.982	0.977*	0.959*	0.952*	0.954*	0.959*
12	3	1.000	1.006*	0.997	0.907*	0.893*	0.899*	0.896*	0.918*	0.892*	0.896*	0.880*
		non- t ext. (Gaussian-GP)										
13	3	–	–	–	–	–	–	–	–	0.960	0.989	0.956
<i>Skew-t K5</i>												
1	10	0.995	1.004	1.015*	1.008	1.010	1.011*	0.993	0.994	0.991	0.992	0.984*
3	10	0.999	1.004	1.006	0.998	1.001	0.997	1.000	0.998	1.003	0.996	1.000
4	3	0.989*	0.993	0.988	0.994	0.992	0.994	0.987	0.985*	0.994	0.988	0.983*
5	3	0.996	0.994	0.992*	0.992	0.994	0.990	0.993	0.997	0.996	0.993	0.997
6	3	0.998	1.001	1.000	0.996	0.983*	0.996	0.986*	0.989	1.002	0.988	1.005
7	3	0.993	0.990	0.997	0.994	0.994	0.994	0.994	0.993	0.995	0.995	1.002
8	3	0.999	0.999	1.007	1.011*	0.997	1.002	1.015*	1.007*	1.005	1.012*	1.008
9	3	1.005	1.007	0.999	0.996	1.007	1.001	1.003	1.003	1.001	1.005	1.003
10	3	0.999	1.002*	1.006*	1.008	1.004	1.005	1.002	0.994	0.988	0.998	0.996
11	3	1.004	1.088*	1.096*	1.123*	1.184*	1.155*	1.142*	1.112	1.073	1.038	1.036
12	3	0.992	1.034	0.995	1.004	0.962	0.976	1.017	1.022	0.965	0.927*	0.913*
		non- t ext. (Gaussian-GP)										
13	3	–	–	–	–	–	–	–	–	1.001	0.977*	0.950*

Table 7: Oracle Brier floor R^* per setting (`oracle_brier_ceiling.R`; Monte Carlo, 3 datasets).

setting	$sd(\bar{\mu})$	R^*	verdict
1 invariant	3.68	0.856	weak headroom (14%)
2 varying	0.42	1.000	dead end
3 ns_dependence	0.41	0.999	dead end
4 ps_cov	0.00	1.000	dead end
5 ps_add	0.39	0.999	dead end
6 covdep_cov	0.00	1.000	dead end
7 covdep_add	0.41	1.000	dead end
8 fuentes_cov	0.00	1.000	dead end
9 gh_marginal	0.43	1.000	dead end
10 lambda_varying	2.80	1.120	harmful
11 invariant_strong	11.04	0.439	positive control
12 mrts_multibump	11.00	0.495	positive control

Table 8: Setting 11, relative score vs K_{MRTS} ($n = 3$): the thresholded symmetric- t methods against the unthresholded skew- t .

score	method	$K=3$	$K=5$	$K=8$	$K=10$	$K=12$	$K=15$	$K=20$	$K=25$	$K=30$	$K=40$	$K=50$
energy	Skew- t K1 (no thr.)	0.999	1.013	1.022	1.027	1.023	1.017	0.996	0.984	0.978	0.966	0.965
	Sym- t K1 (thr.)	0.990	0.263	0.266	0.229	0.201	0.244	0.199	0.256	0.233	0.268	0.305
	Sym- t K5 (thr.)	1.001	0.317	0.539	0.374	0.342	0.462	0.267	0.333	0.359	0.906	0.860
Brier	Skew- t K1 (no thr.)	1.000	1.052	1.074	1.103	1.163	1.138	1.110	1.086	1.048	1.017	1.014
	Sym- t K1 (thr.)	0.999	1.016	1.021	1.039	1.008	0.982	0.977	0.959	0.952	0.954	0.959
	Sym- t K5 (thr.)	0.997	1.078	1.182	1.114	1.163	1.120	1.104	1.133	1.030	1.086	1.063

Table 9: Disambiguation on setting 1 ($n = 10$): relative energy score vs K_{MRTS} , and the *absolute* energy score at $K = 0$. Method pairs (3, 11) and (5, 12) differ *only* in the POT threshold.

	method	$K=5$	$K=10$	$K=15$	$K=20$	$K=25$	abs. ES at $K=0$
3	Sym- t K1, thr. $q(0.80)$	0.630	0.617	0.589	0.579	0.596	21.8
11	Sym- t K1, no threshold	0.996	0.997	0.991	0.990	0.989	8.5
5	Sym- t K5, thr. $q(0.80)$	0.688	1.113	0.773	0.718	0.697	137.1
12	Sym- t K5, no threshold	0.994	1.008	0.993	0.997	0.997	9.3
2	Skew- t K1 (reference)	0.994	0.994	0.985	0.983	0.982	8.5